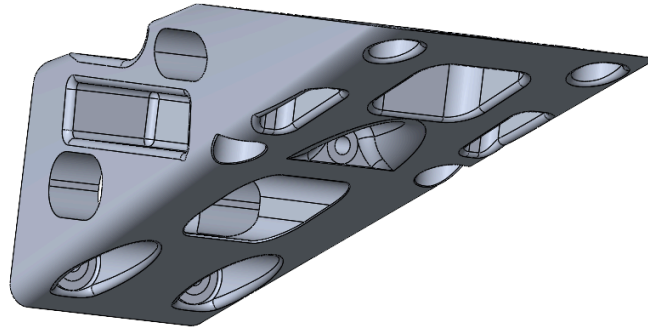


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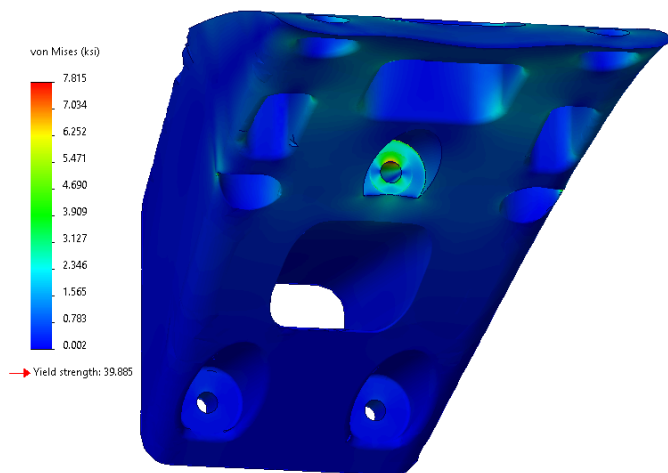
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## CAE Final Project — Optimized Structural Connector

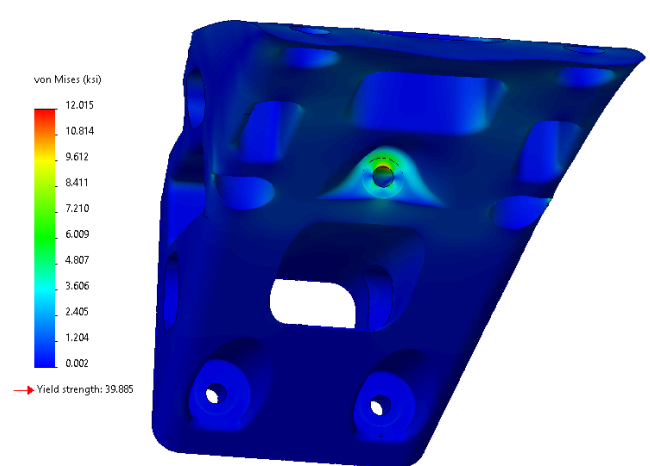
Fully self-modeled in SolidWorks. Brief: bridge two mounting profiles in 6061-T6 aluminum under a 2,000 lbf distributed load, maximizing FOS-to-weight. Five design iterations, each checked with a converged h-adaptive static study. A topology study guided where material came out.



**Final design (Iteration 5 configuration).** Webbed connector with pockets cut where the topology study showed low stress. Fully parametric, with a configuration saved for each iteration.



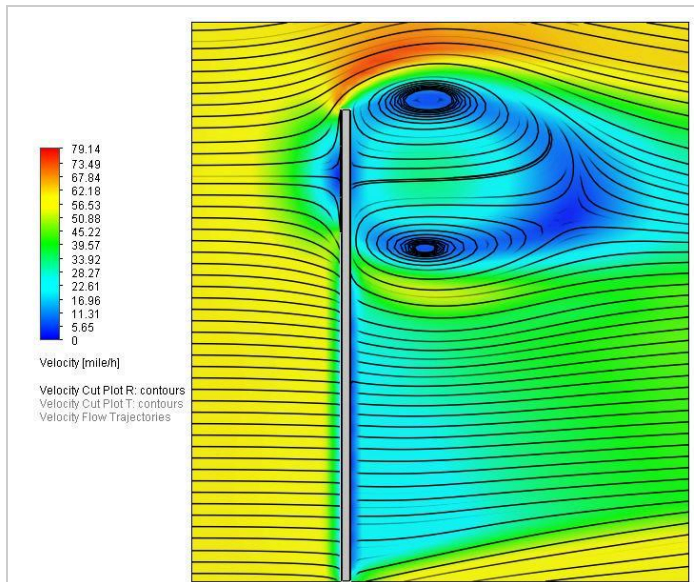
**Static study, earlier iteration.** von Mises stress, max 7.8 ksi against 39.9 ksi yield (6061-T6), converged h-adaptive mesh.



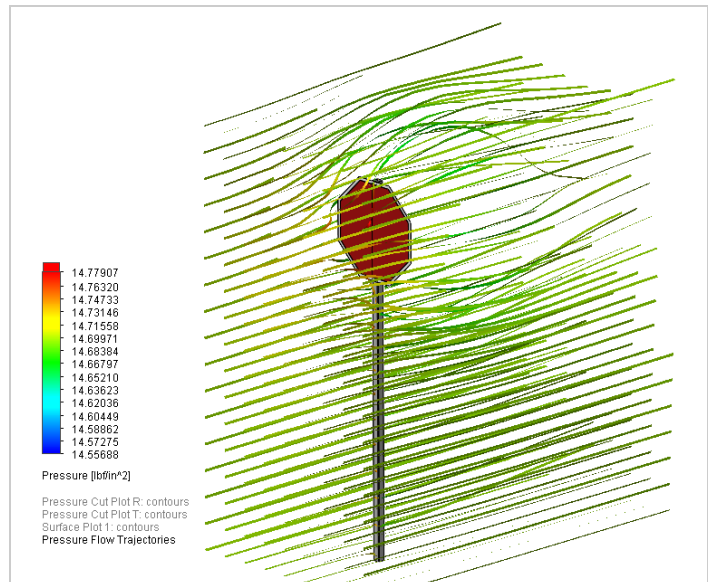
**Static study, later iteration.** After more material came out, peak stress rose to 12.0 ksi. Still well under yield, with less mass.

## Simulation coursework

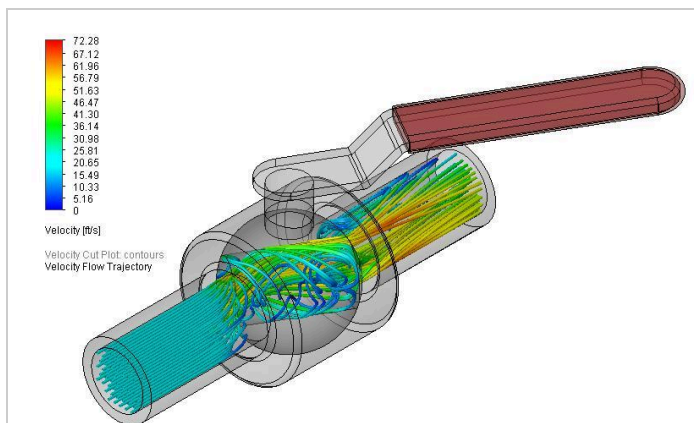
SolidWorks Flow Simulation studies from CAE coursework. Geometry provided by the instructor; flow study setup, meshing, goals, and analysis are my work. Figures below are from my submitted reports.



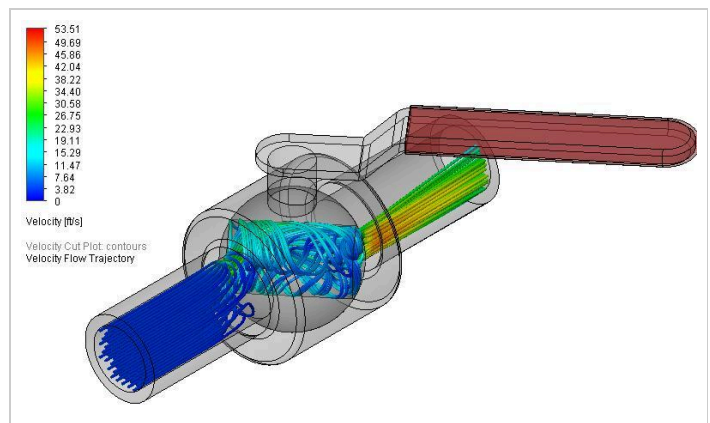
**External flow over a stop sign.** Velocity cut plot with flow trajectories showing wake recirculation and vortex shedding.



**Same study, 3D pressure flow trajectories.** Stagnation pressure on the sign face; the wake forms downstream.



**Internal flow through a ball valve, fully open.** Velocity flow trajectories, peak 72 ft/s through the bore.



**Same valve, throttled.** Flow recirculates in the ball cavity and the jet speeds up through the smaller opening.

*Additional design work: reverse-engineered a Philips water flosser with full drawings and a technical report; designed and built a model car with passive auto-steering including a 3D-printed SolidWorks spoiler.*